Example 3.12 Without actual division , prove that $f(x) = 2x^4 - 6x^3 + 3x^2 + 3x - 2$ is exactly divisible by $x^2 - 3x + 2$

Solution :

Let $f(x) = 2x^4 - 6x^3 + 3x^2 + 3x - 2$

$g(x) = x^2 - 3x + 2$

$= x^2 - 2x - x + 2$

$= x(x - 2) - 1(x - 2)$

$= (x - 2)(x - 1)$

we show that $f(x)$ is exactly divisible by $(x-1)$ and $(x-2)$ using remainder theorem

$f(1) = 2(1)^4 - 6(1)^3 + 3(1)^2 + 3(1) - 2$

$= 2 - 6 + 3 + 3 - 2$

$= 0$

$f(2) = 2(2)^4 - 6(2)^3 + 3(2)^2 + 3(2) - 2$

$= 32 - 48 + 12 + 6 - 2$

$= 0$

$\therefore$ $f(x)$ is exactly divisible by $(x - 1)(x - 2)$

i.e., $f(x)$ is exactly divisible by $x^2 - 3x + 2$

If $p(x)$ is divided by $(x - a)$ with the remainder $p(a) = 0$, then $(x - a)$ is a factor of $p(x)$. Remainder Theorem leads to Factor Theorem.

3.3.1 Factor Theorem

If $p(x)$ is a polynomial of degree $n \geq 1$ and 'a' is any real number then

(i) $p(a) = 0$ implies $(x - a)$ is a factor of $p(x)$.

(ii) $(x - a)$ is a factor of $p(x)$ implies $p(a) = 0$.

Proof

If $p(x)$ is the dividend and $(x - a)$ is a divisor, then by division algorithm we write, $p(x) = (x - a)q(x) + p(a)$ where $q(x)$ is the quotient and $p(a)$ is the remainder.

(i) If $p(a) = 0$, we get $p(x) = (x - a)\,q(x)$ which shows that $(x - a)$ is a factor of $p(x)$.

(ii) Since $(x - a)$ is a factor of $p(x)$, $p(x) = (x - a)g(x)$ for some polynomial $g(x)$. In this case

$p(a) = (a - a)\,g(a)$

$= 0 \times g(a)$

$= 0$

Hence, $p(a) = 0$, when $(x - a)$ is a factor of $p(x)$.

Thinking Corner

For any two integers a $(a \neq 0)$ and b, a divides b if $b = ax$, for some integer x.

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